

Analyte Concentration Determination by Using the External Standard and Standard Addition Methods

Prelab reading: Harris, *Quantitative Chemical Analysis: Chapter 5-3: Standard Addition Method*

1. Learning goals

- Understand the impact of sample matrix on quantitative analyses by comparing the external standard method with the standard addition method.
- Measure uncertainty of slope, intercept, and concentration using statistical tools
- Determine the limit of detection and limit of quantitation using a calibration curve.

2. Reagents

- Reagent grade potassium ferricyanide salt
- Unknown potassium ferricyanide sample

3. Procedures

Instructions for preparation of external standard solutions

- Clean and rinse five 50-mL volumetric flasks carefully with distilled water. A good method to determine if your glassware is truly clean is to observe if any water droplets are present in the neck of the flask. If there are water drops present, the glassware has not been cleaned properly. Perform one final rinse of each flask with the solvent (DI water). If there are not enough 50 mL flasks, make each solution sequentially.
- Dilute the stock solution (≈ 0.02 M) to make 100-mL of a standard addition “spike” solution. The approximate concentration of the spike solution is $5.0 \cdot 10^{-4}$ M.
 - **You do not need more than 15 mL of the stock. Please don't take more than you need.**
- Clean a second beaker with DI water and rinse it with the unknown. Rinsing the beaker with a small amount the unknown will prevent dilution of the unknown. Add ≈ 50 mL of the unknown to the beaker. **You do not need more than 75 mL, so please don't take more than you need.**
- Carefully pipette 10.00 mL of the unknown into each of your five flasks.
- Add the spike solution to each flask with 5.00 mL increments per flask. Flask 1 should have 0.00 mL of the spike, flask 2 should have 5.00 mL of the spike, flask 3 should have 10.00 mL of the spike, etc...
- Fill each flask to the mark with DI water.
- Cover the top of each flask with parafilm and carefully invert each flask ≈ 20 times to mix the solutions well.
- Prepare a table in your notebook that resembles the following in order to keep track of your solutions.

Solution #	Volume of Sample Added, mL	Volume of Standard Added, mL	Volume of Solvent Added, mL	Total Volume of Solution, mL
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Spectrophotometric Measurements

Detailed information on setting up and calibrating the spectrometer is described in Appendix A

- It is good practice to keep all of your results as you collect them, this includes blanks. Keep track of which spectral traces correspond to which solutions in your lab notebook.
- Determine the wavelength of maximum emission for potassium ferricyanide.

- Place a sample of one of your potassium ferricyanide solutions into a cuvette and measure the emission spectrum.
- There should be a large peak 420 nm. Record the location of this peak in your notebook. **You will make all of your measurements at this wavelength during this experiment.**
- Measure the absorbance of the background (DI water) at least ten times at the wavelength that has the maximum absorbance signal.
 - After each measurement discard the solution in the cuvette and replace it with fresh solution.
 - These measurements should be very similar, within 1-2% of each other. If your variability is larger than this, check the calibration settings on the spectrometer.
 - Record each absorbance measurement in a table in your lab notebook.
- Measure the maximum absorbance of each of the standard addition solutions that you prepared three times.
 - After each measurement discard the solution in the cuvette and replace it with fresh solution.
 - Record each absorbance measurement in a table in your lab notebook.
- After you have completed all of your measurements, discard all of your waste in the sink. Clean your glassware with distilled water and return it to the appropriate cupboard.

4. Cleanup

- Clean and rinse all the glassware that you have used and place it back in the cupboard. Place volumetric pipettes in the oven (careful: oven is hot!).
- Wipe your bench space clean.
- Dispose of all other wastes in the designated waste container.

5. Data Analysis and Presentation of Results in Lab Reports.

- There is only one objective for this portion of the lab report: to determine the concentration of potassium ferricyanide in a complex matrix.

Analysis of blank measurements

- Measure the average signal and standard deviation of the ten blank measurements.

Plot of standard addition curve

- *You will need to subtract the signal of the blank from all of your measurements once you've determined the average.*

Sample #, Measurement	Absorbance	Corrected absorbance intensity at max wavelength
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- Plot the corrected absorbance versus $[S]_f$, as presented in Harris Ch. 3-5. If you need help with this, please ask.
- Perform a linear regression of the x,y scatter data and show the trend line on the graph along with the R^2 value.
- Using the LINEST function in Excel, determine the slope, intercept, and uncertainty in the slope and intercept. Report the best fit line along with uncertainty clearly in your lab report.

- Label this as Figure 1 in your lab report, and provide a description of the figure. The figure caption should provide enough detail to understand the result of the experiment.

Calculations

- Calculate the concentration of potassium ferricyanide in the unknown using the standard addition method. Keep in mind that the concentration that you calculate from the graph is the potassium ferricyanide concentration in a 50-mL flask, not in the sample (i.e., you will need to account for the dilution of the unknown sample).
- Present the results in a table similar to that displayed below. Pay attention to units and significant figures.

Concentration of potassium ferricyanide in sample (M)	Standard deviation of concentration (M)	RSD of concentration (%)	Absolute error of concentration (M)	Relative error of concentration (%)
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- Calculate the 95% confidence interval using the method discussed in Ch. 5-3 (note that this is different than the typical way we calculate CI for multiple measurements using a single sample).
- Save your Excel file and upload the Excel file to canvas. Make sure the spreadsheet is well organized and easily understood.

6. Discussion questions

- Briefly describe external standard and standard addition methods. How does it differ from the standard addition method?
- Using the calibration curve from the external standard method, does the calculated value of molar absorptivity agree with the true value at the 95% confidence interval?
- You made two linear regressions (best-fit lines) in this experiment (one for the external standard and another for the standard addition). Using the R^2 values of each line, what can you say about the quality of data and the best fit lines?
- Compare the accuracy of the results for the unknown concentration obtained from the external standard and standard addition methods. Which set of measurements was more accurate? Do these results match your expectations based on your understanding of the external standard and standard addition methods?
- Are the results from either method sufficiently accurate? Assume that a relative error of equal to or less than 5% is good, equal to or less than 10% is acceptable, and larger than 10% is not acceptable.
- Are the results from either measurement precise? Did you expect there to be a difference in precision using the two analysis techniques? Explain your answers.